CMOS INVERTER PROCESS FLOW

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Fabublox process - https://www.fabublox.com/designer/f960771c-00e4-44d2-9cf3-dca13b3b9a24

Vendor Links - https://mtixtl.com/en-internon/products/thermal-oxide-wafer-1000-nm-sio2-layer-on-si-100-4dia-x-0-525-mm-t-ptype-1sp-r-1-20-ohm-cm Starting substrate ligthly p doped Si wafer with 1000nm SiO_2 coating.
https://www.filmtronics.com/capabilities/products/diffusants for the dopants used
https://www.thomassci.com/p/aluminum-etchant-type-a Al type A etchant

I have built a process flow for a Complimentary Metal Oxide Semiconductor (CMOS) inverter as the proposed device for problem A of
the recruitment tasks for HackerFab.
This process flow creates a PMOS(p channel metal oxide transistor) transistor in an n well and a NMOS transistor directly upon the p type substrate, the gates are tied at input and the drains are joined at the output. It uses 6 lithography masks along with multiple steps already within the capabilities of the lab such as BOE etching, sputtering, thermal oxide growth using the tube furnace, and Al wet etching using the Al type A etchant.
The main design decision I made here was to use an aluminium gate instead of a polysilicon gate, as ion implantation or chemical vapor deposition was not feasible. The NMOS reference already used a wet etch Al gate method making it the obvious choice.

HOW IT WORKS:

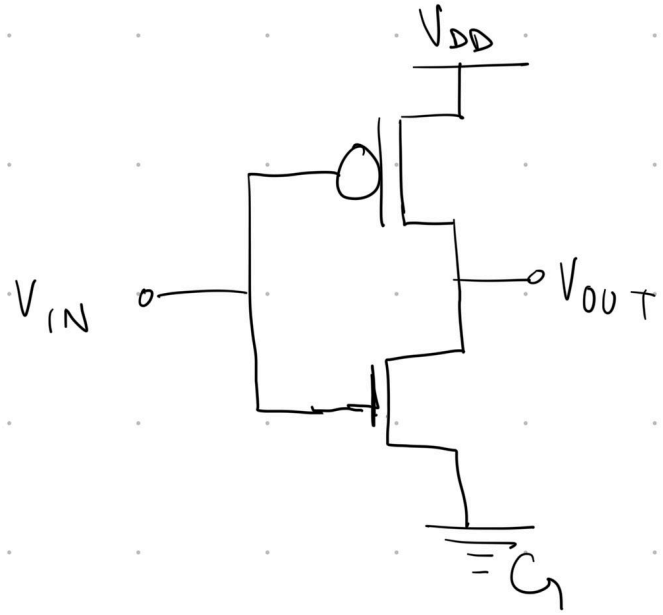
Logic - When Vin is high, the NMOS turns on and the PMOS turns off, and the output is pulled to Gnd. When Vin is low, the PMOS turns on and the NMOS turns off, and the output is pulled to Vdd.

Both transistors are made together on the same wafer. Each mask opens windows over one device and covers the other. The only connections between the two transistors are the metal wires from the last mask.

Part	Location	Made by	Connected to

n-well	Pocket in the p-substrate	Mask 1	Holds the PMOS
NMOS source	N+ region in the substrate	Mask 2	Gnd
NMOS drain	N+ region in the substrate	Mask 2	Vout
NMOS gate	Gate oxide plus Al between its source and drain	Masks 4 and 6	Vin
Substrate tap	P+ region in the substrate	Mask 3	Gnd
PMOS source	P+ region inside the n-well	Mask 3	Vdd
PMOS drain	P+ region inside the n-well	Mask 3	Vout
PMOS gate	Gate oxide plus Al between its source and drain	Masks 4 and 6	Vin
n-well tap	N+ region inside the n-well	Mask 2	Vdd
Contacts	Holes in the oxide over each region above	Mask 5	Lets Al touch the silicon

With one metal layer, this can be laid out without crossing wires: the PMOS sits above the NMOS, one vertical gate strip runs through both, the two drains are joined by a single wire, and the supply rails run along the outside.



V_{IN}	V_{OUT}
HIGH	LOW
LOW	HIGH

Figure 1: Schematic of CMOS inverter

Multiple standard processes were repeated while designing the process which are:

L (lithography)	HMDS prime, spin resist, soft bake, expose, develop	AZ P4210 positive resist, 4000 rpm, 30 s; soft bake 100 C; develop in AZ 400K 3:1 at 25 C for 60 s; exposure time calibrated on the lab tool

W (window etch)	Hard bake, BOE rounds, resist strip	Hard bake 100 C, 90 s; 6:1 BOE at 25 C in rounds until the window is bare silicon (checked by dewetting); strip in acetone then IPA
D (dope and drive)	Spin-on dopant, diffusion, glass strip	Film at 4000 rpm for 20 s; diffusion at 1100 C; 2 BOE rounds to remove the dopant film and the doped surface oxide (had to do 2 mostly for display purposes i.e. clean up the pictures)
O (oxidation)	Thermal oxide in the tube furnace	1100 C, dry O2; thickness set per step
C (Clean)	RCA clean	SC-1, SC-2 and a brief HF dip. Removes organics and residual HMDS before dopant spins and furnace steps

AZ P4210 is a positive resist, so **exposed resist dissolves.**

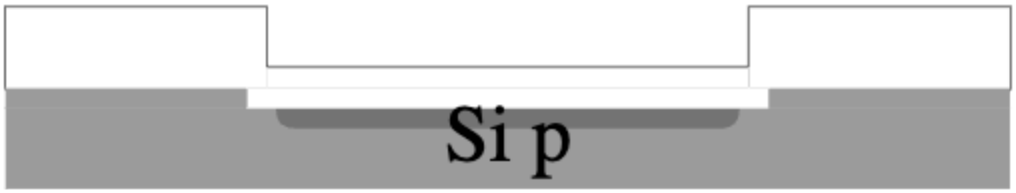
For Masks 1 to 5, the exposed region is the window that gets opened, and the resist stays wherever the oxide must survive. Mask 6 is the reverse: the resist protects the metal that stays, and the exposed aluminum is etched away.

My process has been designed in 7 modules using 6 masks as follows:

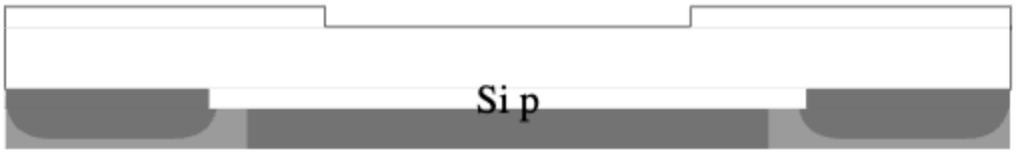
Module	Mask	Sequence	Purpose	Wafer after process
A. Wafer	none	C	Start with the	Si lightly p

			wafer	doped with 1μm oxide coating
B. n well	1	L, W, C, D, O	Open a window, dope with P504, long drive-in, re-grow oxide	n doped pocket in the substrate, covered by oxide
C. N+ source/drain	2	L, W, C, D, O	Open windows away from the well, dope with P504, re-grow oxide	N+ regions for the NMOS and well tap, under oxide
D. P+ source/drain	3	L, W, C, D	Open windows inside the well and at the substrate tap, dope with B155	P+ regions for the PMOS and substrate tap, windows bare
E. Gate Oxide	4	L, W, C, O	Clear oxide over each channel, grow thin dry gate oxide	Thin gate oxide over each channel and the open P+ windows.
F. Contacts	5(2 times)	L, W, C	Open windows down to silicon over every N+ and P+ region	Bare silicon at every contact
G. Metal	6	L, sputter, Al etch	Deposit aluminum, pattern pads, wires and gates	Separated Al features on the oxide and contacts

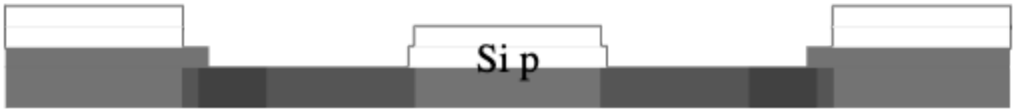
Figure 2: Modules after completion



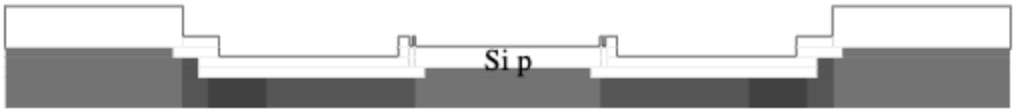
Module B



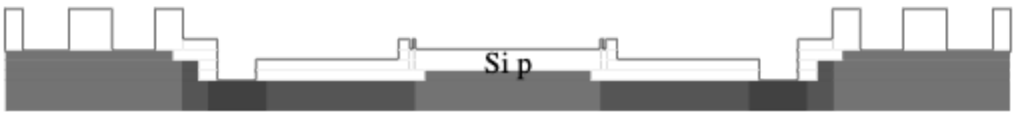
Module C



Module D



Module E



Module F



Module G

Spin-on dopant dopes any bare silicon it touches during diffusion. After each doping step, the opened windows are bare silicon. A thin re-grown oxide covers them, so the next dopant film only reaches the windows opened by the next mask. Without these steps, the boron film would dope the n-well surface and the N+ regions again.

Note: I had to litho and etch twice in module 5 because fabublox behaved unexpectedly and the Al wasnt going all the way to the Si like it should have, the first contact etch did not clear the P+ windows, so I added a second litho round that reopens them. In a real process I would try this as one contact mask, with BOE rounds timed for the thickest oxide, and use the second round only if the thin and thick windows clear unevenly.

The segment ratios for each lithography step are given in the display section of each block in FabuBlox.

Design changes I would implement:

1. The first mask will contain alignment marks, since the wafer is bare at this stage, later masks will align themselves using the XY and θ stages.
2. N+ windows stay about 10 micrometres away from the n-well edge, and P+ windows sit inside the well with a margin, preventing overlap of any regions.

Flaws/Precautions in the current design:

1. The gate window overlaps the edges of the source and drain by more than the alignment error. The gate is not self-aligned, so this overlap adds capacitance.
2. Gaps between metal features must be larger than the lab's 4.7 micrometre resolution.

Differences between FabuBlox process and practical flow:

1. The editor drew the diffused layer as a thin region, and the re-oxidation consumed it. A real well will go several micrometres deep, and an oxidation of about 200 nm will consume only about 90 nm of silicon at the surface (~44% of the oxide thickness)
2. I had to add extra etch blocks just to fix the display parts and to get the final result I desired, this won't be the case in the lab. I had to use the extra blocks to get rid of the doped SiO_2 glass, HMDS and others which will be removed in one etch only.
3. This is only a cross sectional view and does not show the entire topology.
4. The display elements have not come across as well as I would've liked them to.

PRACTICAL CHALLENGES:

1. **N well is too shallow** - the phosphorous moves into the Si at a very slow rate (few micrometers/hour) and every time an oxidation takes place it eats into the thin doped region, if the n well is too shallow, the PMOS wont function properly
    a. Solution - Increase doping diffusion time at 1100ºC, use test wafer and a 4 point probe to check for sheet resistance(lower resistance implies more doping), make P+ doping shorter so that the level stays above the n well
2. **BOE** is a hydrofluoric acid mix. It gets under the edge of the resist and peels it, so **windows come out ragged** or in the wrong place.
    a. Solution - remove all surface water before priming, spin resist immediately after HMDS step, soft bake first and then hard bake after developing the resist. Etch in short rounds and use a microscope to observe after each round, using water to check if it beads up. As a failsafe we can always put down a chromium layer as a mask, pattern it, etch the oxide through it, and then remove it.
3. **Misalignment Errors** - Since i am using 6 masks the alignments errors if not corrected for can add up resulting in overlapping regions of the PMOS and NMOS resulting in shorts and broken circuits.
    a. Solution - Alignment marks on the first mask. Every later layer lines up against the same marks, so errors don’t pile up. The XY and θ(rotation) alignment should be ensured to be correct before exposing. Keeping margins larger than the possible shift if the lithography machine will also help reduce the error. However, the bigger margins and overlap error add capacitance and make the circuit slower.
4. **Poor gate oxide growth** - the oxide growth can lead to multiple problems within the circuit. Possible defects include:

Problem	What happens	Effect	Solution
Wrong thickness	Oxide layer is thicker or thinner than needed	V_{th} changes	Grow test oxides first, keep furnace settings fixed, and confirm thickness with MOSCAPs.
Mobile ions	Sodium or other metal ions trapped in the oxide that drift when voltage is applied	V_{th} drifts with time and voltage	RCA clean, clean handling and tube. Test with a heat-and-bias C–V check.
Trapped charge and interface defects	Charge in the oxide or at the silicon surface	V_{th} shifts, and the transistor conducts less current	Use <100> silicon, anneal in N_2 after oxidation, and anneal at about 450 °C after the metal.
Pinholes or weak spots	Tiny defects in the film	The gate leaks to the channel, or the oxide breaks down at a voltage below V_{DD}	Keep the wafer particle-free, RCA clean right before the furnace, and use dry O_2. Check leakage and breakdown on several MOSCAPs.
Dopant contamination	Boron or phosphorus from the shared furnace or from leftover film	Local doping changes, so V_{th} shifts and varies between devices	Strip dopant glass completely (boron glass especially), use separate boats and a baked

			tube, and run the gate oxide before dopant runs. Check sheet resistance on a control piece.

Also:

a. The silicon surface has been through a lot of different processes before the oxide grows. By Module E the channel region has been through BOE rounds, resist and HMDS, and boron and phosphorus films. Any leftover residue or doped skin ends up under the oxide.
b. Native oxide and cleaning. If the wafer sits in air after the clean, a thin native oxide grows on it and the clean's benefit is lost before the furnace.

5. **Poor contacts to the silicon** - a thin natural oxide on exposed silicon blocks the metal, and the contact becomes resistive. The thick and thin windows also etch at different times. This can be tested for by checking current through a contact test structure which should give a I-V characteristic similar to an ohmic conductor.

Solution - Etch until the thickest oxide disappears since BOE doesn't affect the Si. Before loading onto the sputter, dip the wafer in dilute HF for a few seconds to remove the natural oxide, and load it into the sputter chamber quickly. After the metal is patterned, bake at about 400 to 450°C in nitrogen to ensure that the Al bonds to the Si. However, Al and Si start to alloy above 577ºC so stay below that temperature.

6. **The gate overlaps the source and drain** - As the gate isn't self aligned, it has to overlap both sides, and that overlap acts like a small capacitor that slows the circuit. This results in the inverter switching slower than usual. This is a trade-off we need to make since self aligned polysilicon gates need CVD.

I would not run the inverter first. On the same wafer, I would include

1. MOSCAPs for gate oxide thickness, fixed charge, mobile ions and leakage,

2. diodes and sheet-resistance patterns for the well, N+ and P+ doping,

3. contact test structures (two pads on one doped region, swept in I–V, with several spacings for TLM (Transfer Length Method),

4. single NMOS and PMOS transistors for threshold voltage and drive current, and finally

5. the inverter’s voltage transfer curve. Test structures check that the process works, and the single-device and inverter measurements check the actual devices.

Eventually, I plan on joining multiple of these CMOS inverters to create a ring inverter.